Data Structures and Algorithms · 21CSC201J

Unit 3 · Session 11 · SLO 1 — Solved Worksheet

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Unit â€“ 3 : Session â€“ 11 : SLO â€“ 1 | Circular Queue â€” Introduction

I. Fill in the Blanks

• A circular queue connects the LAST position back to the front.
• Circular queue overcomes the problem of MEMORY WASTAGE (unused spaces) in a linear queue.
• In a circular queue, when the rear reaches the end, it wraps around to index 0.
• A circular queue follows the FIFO principle.
• The condition for a full circular queue is: (rear + 1) % size == FRONT.
• The condition for an empty circular queue is: front == -1.
• When the first element is inserted, both front and rear point to the SAME (0th) index.
• Circular queues are commonly used in ROUND-ROBIN scheduling.
• The ISFULL function checks if a circular queue is full.
• The ISEMPTY function checks if a circular queue is empty.
• Circular queues help in utilizing memory more EFFICIENTLY than linear queues.
• In a circular queue, incrementing the rear is done using MODULO (%) operator.
• Circular queues can be implemented using arrays or LINKED LISTS.
• The front element of a circular queue is accessed using the FRONT pointer.
• In circular queues, rear and front move in a CIRCULAR (wrap-around) manner.